put back in a locomotive boiler. Two and one-fourth inch flues weighing less than 2 pounds to the foot are not put back in a locomotive boiler. The best of these flues are assorted and used in stationary or pump boilers.

Our next operation is to cut the flue to length for welding. This is an extra operation, due to the fact that we weld on both ends of our flues alternately, and to avoid having a weld come too close to the front flue sheet, which might bother when rolling same, we find it absolutely necessary to cut flue to length for welding.

We cut our steel flues up in pieces 5 inches to 7 inches, 8 inches or 9 inches, as the requirements may be, for welding on or safe ending. These pieces are cut off on a machine we have for that purpose, which cuts off the piece, bevels and removes the burr at the outer end in one and the same operation.

We believe in taking the burr off of the inside of the safe-end piece at the firebox end. This so that flue will pass over flue mandrel on welding machine readily. By removing this burr it also allows us to use a larger mandrel on flue welding machine. This helps us to keep the flue the original size.

Many roads do not take time to remove this burr or use the larger mandrel, but their flue shows to be contracted at each weld. We are using for a 2-inch flue a 1 11/16-inch mandrel, which is about the original inside diameter of flue.

After the flue is cut to length we heat and open up all flues at front end to fit flue hole in front flue sheet. This to avoid shimming, which takes time and is poor practice.

Many roads are still welding on but one end of the flue, in some cases as many as five and six welds, and every weld on a flue is one more chance for a failure.

Many times flues are slightly overheated at weld and become quite thin just back of the weld. Leaks do develop at these points and flues do break in two at these welds, sometimes in the rattler and other times while in service. So I say the less welds you have on a flue the better.

A flue with five or six welds at one end might weigh up all right, but the old part or the original would not be good for further service.

As I stated before, our practice is to weld on both ends of the flue alternately, and flue put in service to-day with safe end at firebox end. This flue may run from two to three years, and when removed the bead is cut off at firebox end, and at the front end with flue cutter about 1½ inches from end of flue.

Now before welding this flue we cut to length for welding and apply the safe end piece on the opposite end of flue. Then, when flue is welded, we cut to proper length to go in the boiler, but only have to cut 1 inch or 1½ inches off of the original safe end.

This flue again goes into service for another two or three years, and when flues are taken out bead is cut off at firebox end and at front end with flue cutter as before, is then cleaned and cut for welding. This time we cut the original weld off and the new piece or safe end is again welded on the front end, which leaves the flue with but two welds, and this operation continues throughout the life of the flue. It always gives us a good new safe end at firebox, the old safe end at front flue sheet.

By following this method we don't know what it is to split a flue in the front end or to damage a flue in the rattler and we have a better flue at all times than where safe end has been repeatedly welded on one end.

A great many might object to the cutting of this flue for welding, but unless you do this you cannot tell where the old weld will come at the front flue sheet.

The cost for cutting flues to length for welding is less than ½ cent per flue, or about 50 cents per 100. I believe if this practice was carried out you would have a flue that is worth that much more to you at all times.—*Railway Master Mechanic.*

Advantages of Mechanical Stokers*

The advantage to be derived from stoker firing of locomotives is the ability to fire the engine continually up to its capacity, and it is found that the stoker-fired locomotive can either take the same tonnage as the non-stoker over the road in less time, or a larger train can be handled in the same time. As a concrete example of this fact we note that in a recent test for the capacity of locomotives a stoker-fired engine was operated for six hours, firing an average of 7,800 pounds of coal per hour, which means a continual capacity of the locomotive firing in excess of that which could be maintained by hand firing.

It has also been demonstrated that mechanical stokers have permitted the enlarging of the exhaust nozzle area from 5 to 5¼ inches, which means an increase of about 3 square inches, thus giving the locomotive greater efficiency. Other advantages are obtained by not opening the fire doors, viz.: doing away with the glare or dazzling light which is produced after dark and which makes the observation of signals more difficult. It also prevents sudden change in firebox temperature, which produces contraction of sheets or tubes.

A properly adjusted mechanical stoker will reduce the use of fire hook or rake on the fire bed, as the distribution of coal can be regulated to prevent banking. This is an advantage, as the frequent use of the rake disturbs particles of fuel which are carried by the draft onto the brick arch or lodged in the flues, reducing the draft.

The application of the stoker has proved to be a benefit from the standpoint of smoke abatement, and there are some stoker locomotives at present being used in the heavy transfer service within the limits of large cities, resulting in practically complete elimination of smoke. Although all types of stokers are not showing an improvement in smoke prevention, the good results of some types indicate that future development may be expected to produce good results along this line.

Those engaged in stoker firing do not have to devote as much time and attention to the use of the mehods employed in hand firing, but are required to operate the mechanical stoking machine, which furnishes them with a practical experience in the care of steam-driven machinery. This mechanical education should greatly aid in the development from fireman to locomotive engineer. This, we believe, is an improved condition and should greatly increase the possibilities of securing a higher type of candidate for the position of locomotive engineer.

We have not received any information which indicates that the development of the mechanical stoker has reached a point where the utilization of cheaper fuel has been accomplished.

In conclusion, we will not attempt to state the cost of installation and maintenance of the mechanical stoker, as the different types will vary in these items, but from the developments up to date it is safe to say that the advantages may be expected to greatly multiply with the service of the machines and by the efforts of the various stoker manufacturers.

* Report presented before the Traveling Engineers' Association.

this matter that the staybolts are given an angular or lateral pull by these movements, and the most severe of these are when there are the greatest temperature differences within the firebox.

These extreme differences occur in hasty firing up, cooling down, cold water washing or filling, and from cold drafts after fires are knocked out, and in most of these cases there is no pull on the bolts from much, if any pressure, nor is there water circulation. When the boiler is in normal operation, it is doubtful if the angular strains are excessive, and it is probable that the initial rupture, the first start of the breakage, occurs at some firing up time or other period of extreme temperature variation.

Staybolts break under the pull due to pressure, but only after the cross section has been reduced by checking or partial fracture induced by over-stressing of the outer fiber at times of extreme movements, which are at times when there is probably little or no pressure.

The larger the bolt beyond actual requirements, the more probability of its early failure, for the obvious reason that the stiffness and rigidity increases with the size, and large bolts are unable to respond to the firebox movements very many times before the fracture starts which winds up with a broken bolt.

Almost invariably solid bolts break close to the outer sheet, and due to this the practice of drilling a telltale hole in the outer end gives a warning of a broken or fractured bolt. The reason for this location of breakage is because the outer sheet is always heavier than the inner, is not as subject to movement and acts as a foundation for the staybolt, the other end of which is pulled laterally by the firebox movement.

The comparative flexibility of various sizes of solid staybolts can be approximated mathematically, and also roughly, by vibratory machinery, although no machine has yet been devised to give accurate comparison. This is perhaps also due to lack of complete knowledge of the laws governing the effects of repeated stresses on materials. At all event, the vibratory test is not yet in good repute among engineers, and this puts an uncertainty on determining the real merits of staybolts by test other than service, save what can be deduced from the results of pulling tests in the elongation and reduction and in nicking and bending tests.

It is now clear that flexibility consistent with strength is a desirable and necessary feature of locomotive staybolts, and that increase of size and strength decreases the more vital element of flexibility upon which largely the life of the staybolt depends. The average performance has already been stated, and it is apparent that the needful and necessary thing is to increase the body flexibility to withstand without failure the vibrations that may occur during the life of the firebox at least. Much more than that is not necessary, nor justified, at least so far as the staybolts themselves are concerned.

There is another feature, however, of prime importance, interdependent with staybolt life and flexibility—i. e., firebox life. The effect of stiff, rigid body staybolts on the sheets has not been generally appreciated as to their destructive action. To give perfect freedom for parallel movement of the inner and outer sheet consequent on unequal expansion and contraction, it is evident that the staybolts should be provided with the equivalent of a universal hinged joint at each end or that the body should be so flexible as to yield the necessary flexure. If the bolt body is stronger than the sheet then the angular movement of the bolts tends to buckle the sheet. As the outer sheet, on account of its thickness, is generally stronger than the bolt, it does not buckle, but the bolt bends close to the sheet, and if the bending overstresses the outer fiber it starts the check or initial fracture then and there. These, oft-repeated, account for broken staybolts, but what about the other end? The inner sheet being lighter is not as strong as the staybolt and yields by buckling around the hole to the angular movement of the bolt. Every boiler maker is familiar with the worked appearance of the sheet around staybolt holes in sheets which have been renewed. The bolts act as levers in the holes, stressing the material beyond the yield point, causing leakage, and this many times repeated leads to failure induced by radiating cracks. The extension of these cracks means firebox failure; their prevention means a measure of firebox life, as fireboxes seldom fail without premonitory cracks. The cracks as a rule start at the holes, and the evidence seems to be against the leverage action of the staybolts as being responsible. Here again flexibility seems to be the quality necessary to effect a reform that will add life to fireboxes as well as to staybolts. The comparative value of the two is manifestly in favor of the firebox. It costs about $1,000 to renew the box of an average size of engine, without counting the value of the loss of its time, and the addition of a year or two or more is well worth while. Stiff rigid body staybolts may be used with apparent good life, but at the expense of the flexibility of the firebox sheets, and it is not a hard question to solve as to whether the box does not cost much more in the end than the staybolts. The life of the two should be as coincident as possible, there being no salvage except scrap value of staybolts when the firebox is renewed.

As additional flexibility of the staybolt body seems most desirable to conserve firebox and staybolt life, a little study of

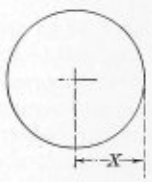


Fig. 1.

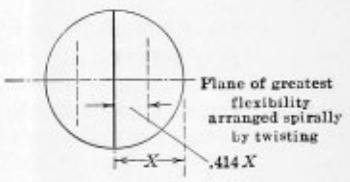


Fig. 2

the situation and possibilities will yield good economic results. From the foregoing arguments it is evident that large stiff body staybolts are a mistake—a costly one—as they fail because of their own rigidity and lack of flexibility to respond freely to the lateral movements required, and they also buckle and work the firebox sheets, inducing cracks and early failure. It would seem, therefore, that staybolts as small as a sufficient factor of safety will permit should be used and maintained so far as possible in firebox renewals.

Additional flexibility which will increase the staybolt and firebox life can be obtained by further sub-dividing the body of the staybolt, and it has been said that a rope or cable would make an ideal staybolt. The principle is true, but the practical application would be difficult on account of many factors. Manufacture of such a product must turn out a structure that will not greatly exceed the cost of the common staybolt; must have and maintain threads in continuous pitch during its application; must take the telltale hole required by law and also indicate the staybolt condition by the hammer test; must not too greatly multiply surface that corrosive waters can act upon to its early destruction.

These features have all been duly considered and the results are exemplified in the American Staybolt, which has a two-piece body, and this is twisted so that its most favorable plane of flexibility is presented in every direction. Fig. 1, below, represents the cross section of a solid staybolt body of approximately ⅞-inch minimum diameter. The relative flexibility of this size as compared with any other size, measured in outer fiber stress, is in proportion to the respective radii, this distance being shown by "X," which is the distance from the center or neutral axis to the outer fiber. This statement is true provided the sheets do not buckle and thereby relieve the bolt to some extent. The American Staybolt body cross section, as shown by Fig. 2, consists of two half rounds. Now

excessive bending stresses, set up by the unequal expansion and contraction of the inner and outer sheets. The load upon a staybolt, due to the boiler pressure, is comparatively small as compared to the stress induced by the inequality of expansion and contraction.

STRENGTH OF STAYBOLTS

The strength of a staybolt should be calculated from its smallest area. Staybolts are made from wrought iron on account of its fibrous structure, which will stand more abuse from the different stresses acting upon the bolt than will steel, which is of a crystalline structure. All staybolts become more or less crystallized by the rapid blows of the hammer in riveting, although it is impossible to judge the amount of crystallization occasioned by the riveting process.

Inferior installation of flexible staybolts often gives us considerable trouble. We have been calking the flexible bolt thimbles or bushings in the roundhouse for the last three years, and, furthermore, we have not finished calking yet, and probably will not have finished until we have gone over the entire lot of bolts. We may attribute this defect to inferior installation during the construction of the boiler in the locomotive works. This is an expensive item for our railroads to contend with, and it should be eliminated.

APPLICATION OF FLEXIBLE BOLTS

When applying flexible staybolts, the adjustment of the bolts should be taken into consideration, although this is not uniform for all types of boilers. When firing up a locomotive boiler, the inner firebox sheets expand more rapidly in an upward diagonal direction than the outer sheets. As steam is raised to its working pressure the outer shell expands in a direction which extends longitudinally to a greater extent owing to its larger dimensions and greater thickness. The difference in the amount of expansion between the sheets varies in different types of boilers and, for this reason, bolt adjustment should be based upon data obtained by tramming the firebox for the difference in sheet expansion. However, the writer is of the opinion that the following course of adjustment is adaptable for fireboxes 8 feet long and under:

ADJUSTMENT FOR SIDE SHEET AND BACK HEADS

Taking into consideration the first, second and third outside rows and the same across the top, the bolts in row No. 1 should be given a half turn back; those in row No. 2, three-eighths of a turn back, and those in row No. 3 a quarter of a turn back. For fireboxes 8 feet in length and over, the bolts in row No. 1 should be given three-quarters of a turn back; row No. 2, one-half of a turn back; row No. 3, three-eighths of a turn back.

For the adjustment of the throat sheet, taking the first three rows above the mudring, the first row above the mudring should be tight; the second row, three-eighths of a turn back; all other rows, three-quarters of a turn back.

When large areas are covered by flexible staybolts, all bolts inside of the three outside and the three top rows should be turned back off of their seats one-eighth of a turn, because riveting has a tendency to draw the bolt up to its seat.

Staybolts should have a larger factor of safety than the boiler shell or plates, on account of its being subject to both a direct and an indirect pull, as well as an unequal vibratory stress. For this reason I do not think it is practical to admit staybolts in the rivet line when applying a patch to firebox sheets.

EXAMPLES

What load is carried by a staybolt when the bolts are spaced 4 inches between centers and there is a steam pressure of 150 pounds per square inch? The load carried by the staybolt is equal to the area it supports multiplied by the steam pressure per square inch, which, in this case, will be $4 \times 4 \times 150 = 2{,}400$ pounds. This is, of course, disregarding the area of the bolt itself.

If we allow 6,000 pounds stress per square inch for staybolts, what area would a staybolt support, the least diameter of the bolt being ⅞ inch and the allowable working pressure 200 pounds per square inch? First determine the area of the staybolt as follows:

$$.7854 \ (.875)^2 = .6013.$$

Then multiply the area of the staybolt by the allowable stress per square inch, and divide the result by the allowable working pressure, giving the area as follows:

$$\frac{.6013 \times 6{,}000}{200} = 18.03.$$

Finally extract the square root of the quantity representing the area, and you will have the spacing or pitch of the staybolts:

$$\sqrt{18.03} = 4.25, \text{ or } 4\tfrac{1}{4} \text{ inches.}$$

What force will a staybolt resist whose smallest diameter is ¾ inch, the diameter at the root of the thread being ⅞ inch, with a 3/16-inch telltale hole, and the allowable working stress 6,000 pounds per square inch? First the area at the root of the threads must be determined. From this value deduct the area of 3/16-inch telltale hole, or $(.875)^2 \times .7854 - (.1875)^2 \times .7854 = .5737$ square inch.

The area of the staybolt at its ¾-inch diameter will be $(.75)^2 \times .7854 = .4418$ square inch. The area for the ¾-inch diameter being less than the area at the root of the thread minus the area of the telltale hole, the load allowed is computed from the ¾-inch diameter and is: $.4418 \times 6{,}000 = 2{,}650$ pounds.

THE HAMMER TEST

Boilers are sometimes put under pressures of from 40 to 50 pounds per square inch to aid the inspector in locating a broken staybolt with the hammer. Putting the pressure on the boiler causes the two parts of the broken staybolt to separate, thus permitting the broken staybolt to be more readily found.

All staybolts, crown-bolts and radial bolts should be placed so as to be at right angles or 90 degrees to the sheet they support. When the pitch of the staybolts is excessive, the pressure will bulge the sheets and thus create a deformation.

Staybolts should project beyond the sheet about two threads to form a head in driving the bolt. Excessive allowances make it difficult to upset the staybolt in the hole. The smallest size staybolt advisable for high-pressure boilers is ⅞-inch diameter.

In the distribution of staybolts and braces, every effort should be put forth to distribute them so that each staybolt or brace will have the same working stress per square inch; that is, as nearly so as practicable. In arranging staybolts, attention must be paid to the size of the staybolt, the pitch and the thickness of the plate it supports. It may be possible that the staybolt or brace will be large enough to support the area allotted to it, but the plate may be so light that the pitch will be excessive and cause deformation of the plate. In deciding upon the pitch it is necessary to know the formula generally adopted for this purpose.